

Mohammad Nusairat

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EDUCATION

University of Illinois at Chicago

Chicago, IL

B.S. in Data Science with a Concentration in Computer Science, GPA: 3.50

May 2025

- Related Coursework: Data Structures & Algorithms, Advanced Data Structure Practicum, Intro to Machine Learning, Software Design, Database Systems, Optimization for Analytics, Business Project Management

SKILLS

Programming Languages: Python, Javascript, TypeScript, C++, C, C#, Java, SQL, R, SAS

Technologies: Flask, React, Next.js, Node.js, Tailwind CSS, Railway, Git, VS Code, Jupyter Notebook, Anaconda

EXPERIENCE

Comprehension, Collaboration and Creativity Lab

University of Illinois at Chicago, IL

Undergraduate Research Assistant

Sept 2024 - May 2025

- Research cognitive patterns in **algorithmic** problem-solving for a study on “Aha! moments” in programming
- Analyze participant interviews, **annotated** insight-related data, **co-authoring** findings **presented** at MPA 2025

Outlier

Remote

Software Developer - AI Trainer

July 2024 - April 2025

- Evaluate the quality of AI-generated Python/C++ code by developing **robust test cases** to train the model
- Enhance the accuracy of LLM’s code through **data ingestion** by solving complex algorithmic problems

Headstarter AI

Remote

Software Engineering Fellow

July 2024 - August 2024

- Built 5 **AI apps** & APIs using NextJs, OpenAI, Pinecone, StripeAPI with 98% accuracy as seen by 1000 users
- Developed projects from design to deployment **leading 4 engineering fellows** using **MVC** design patterns
- Coached by Amazon, Bloomberg, and Capital One engineers on **Agile**, **CI/CD**, **Git** and **microservice** patterns

PROJECTS

Arabic YouTube Transcript Translator

[Website](#)

- Built a **React + Vite** frontend and **Dockerized Node.js + Express.js** backend web app that transcribes Arabic YouTube videos, translates them into English, and generates timestamped PDF, Markdown, and SRT files
- Streamlined content creation for my personal business by **reducing Arabic content preparation time by 86.7%** and **increasing transcription and translation accuracy by 20%** as tracked by internal QA logs
- Achieved **50k+ LinkedIn** views with **330+ users** and **16+ hours transcribed** on launch day

Campus Meeting Point Application

[Website](#)

- Engineered a full-stack web app **significantly reducing** meeting problems on UIC’s undergraduate campus
- Built using **Flask** and **React**, the app uses **XML parsing of OpenStreetMap data** to calculate the optimal meeting point between any number of users on campus, filtered by user-selected building utilities and features
- Developed scalable **REST API** endpoints supporting building autocomplete, tag-based filtering, and geospatial pathfinding using **Dijkstra’s algorithm** and **Geodesic Fermat point logic**; deployed via Railway and Vercel
- Engineered fault-tolerant location logic and dynamic frontend components for search, filters, and ETA summaries, enabling real-time user coordination with location-aware filtering via **NetworkX** and **geopy**

Book Review Application

[Website](#)

- Built a full-stack web app with **user auth**, **search**, **review features**, and Railway MySQL **cloud deployment**
- Integrated **advanced SQL logic** like search filters and average rating via GROUP BY, LIKE, ORDER BY
- Seeded **100+ book records** using Python + Pandas; enabled dynamic CSV-to-SQL loading with error handling
- Built secure login/registration with **SHA-256 hashing** and session-based access control
- Implemented **dynamic search** + review UX with Flask, Jinja templates, and modular route handling

PUBLICATIONS

Rahman, J., Virruso, A., Nusairat, M., Khan, R., Miller, T. S., & Wiley, J. (2025, April 10-12). Examining the Aha! experience in computer programming. Poster presented at the 97th Annual Meeting of the Midwestern Psychological Association, Chicago, IL